

# Phase 8I — Unified Risk API & Operational Prediction Contract

Technical Walkthrough, FastAPI REST Service & Pydantic V2 Architecture

|                                                             |                                                         |
|-------------------------------------------------------------|---------------------------------------------------------|
| <b>API Framework:</b> FastAPI 0.135+ / Pydantic V2          | <b>Host/Port:</b> 0.0.0.0:8000                          |
| <b>Endpoints:</b> /health, /info, /predict, /profile        | <b>CORS Policy:</b> Whitelisted origins (No wildcard *) |
| <b>Error Sanitization:</b> Zero stack traces / paths leaked | <b>Test Suite:</b> 142 / 142 Tests Passed               |

## 1. Primary Endpoint Contracts

| Endpoint        | Method | Response Model  | Latency (p50) | Key Functionality                                     |
|-----------------|--------|-----------------|---------------|-------------------------------------------------------|
| /api/v1/health  | GET    | HealthResponse  | 9.59 ms       | Lightweight liveness probe (Zero raster I/O)          |
| /api/v1/info    | GET    | InfoResponse    | 12.9 ms       | Model metadata, features, thresholds, disclaimers     |
| /api/v1/predict | POST   | PredictResponse | 316.0 ms      | Unified prediction (Static LSM + CWC telemetry + IMD) |
| /api/v1/profile | POST   | ProfileResponse | 294.5 ms      | Static-only location profiling without rainfall       |

## 2. Timezone & Security Guardrails

- **Timezone Enforcement:** ISO-8601 strictly validated. Timezone-naive timestamps rejected with HTTP 422 NAIVE\_TIMESTAMP\_REJECTED. Omitted timestamps safely default to current UTC.
- **Error Normalization:** Structured JSON error contract with stable codes (OUTSIDE\_SUPPORTED\_DOMAIN, INVALID\_COORDINATES, etc.). All stack traces intercepted.
- **Baseline Integrity:** All 10 baseline assets verified bitwise identical.